

AARON JALAPON ARQUILLANO

SOFTWARE & ML ENGINEER | PRODUCT-MINDED DEVELOPER

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PROFESSIONAL SUMMARY

Product-minded Software & ML Engineer who builds full-stack applications, machine-learning systems, and user-centered product prototypes across agritech, smart mobility, computer vision, generative AI, and engineering simulation. Experienced with Python/FastAPI, React/Next.js, PyTorch/scikit-learn, data pipelines, model evaluation, APIs, and deployment tooling.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, PHP, Dart

Frontend: React, Next.js, Streamlit, Flutter, D3.js, Recharts, Plotly

Backend & APIs: FastAPI, Node.js, Laravel, REST APIs, WebSockets, MQTT

ML & AI: PyTorch, scikit-learn, XGBoost, LightGBM, Prophet, Faster R-CNN, VGG16, RAG, LangGraph, ChromaDB

Data & Databases: PostgreSQL, MySQL, MongoDB, time-series data, feature engineering, model evaluation

Cloud & Tools: Docker, Google Cloud Run, Google Cloud TTS, Git/GitHub, pytest

SELECTED ENGINEERING PROJECTS

TanUM Nanotech Solutions | *Software & ML Lead* | *Active Pilot*

Aug 2025 - Present

Python, MQTT/HTTP, Machine Learning, React, Time-Series Data, Docker

- Designed cloud ingestion, data cleaning, and time-series storage for NPK, pH, moisture, EC, and temperature readings streamed from IoT soil sensors.
- Implemented the fertilizer-recommendation pipeline, API/dashboard integration, and low-bandwidth SMS alerts for farmer decision support.
- Defined pilot targets of 20-30% fertilizer reduction and estimated 15-25% input-cost savings; results remain targets pending broader field validation.

IQueue | *Full-stack Developer / AI-ML Systems Developer* | *Hackathon Prototype*

May - Jun 2026

FastAPI, Next.js, PostgreSQL, Prophet, PyTorch, LightGBM, XLM-RoBERTa, QR Security

- Built passenger and operator flows plus backend services for seven-day demand forecasting, surge recommendations, explainable seat assignment, and evidence tracking.
- Implemented HMAC-SHA256 signed QR boarding passes for offline validation and tamper rejection.
- Fine-tuned a multilingual intent classifier for English, Filipino, Bahasa Indonesia, and Vietnamese, reaching 0.8443 holdout accuracy on 244 examples; operational evidence is synthetic/demo-based.

Banana Grading System | *Sole Developer / ML Engineer* | *Academic Prototype*

Jan - Mar 2026

Python, PyTorch, Faster R-CNN, VGG16, Streamlit, COCO Evaluation

- Trained a Faster R-CNN ResNet50-FPN-v2 detector that achieved 0.9675 mAP@0.50:0.95 and detected bananas in all 54 held-out test images.
- Built VGG16 transfer-learning classifiers for maturity, quality, size, and front/back view, with best-method held-out accuracy ranging from 82.5% to 97.8%.
- Shipped a local Streamlit workflow with annotated detections, confidence scores, ensemble/TTA controls, evaluation artifacts, and automated tests.

podcastman | *Full-stack AI Engineer / Solo Developer* | *Deployment-ready MVP*

Feb - Mar 2026

FastAPI, Streamlit, LangGraph, LiteLLM, ChromaDB, Google Cloud TTS, Docker

- Built a URL/text/Markdown ingestion pipeline that normalizes and chunks sources, generates embeddings, and retrieves grounded context from ChromaDB.
- Orchestrated generation, accuracy checking, storytelling, and engagement stages with LangGraph to produce two-host podcast scripts.
- Implemented dual-voice TTS, audio normalization and assembly, tests, Dockerfiles, and a Google Cloud Run deployment script.

SELECTED ENGINEERING PROJECTS - CONTINUED

AirFlow Sim | *Solo Developer / Simulation Engineer | Local/Docker Demo*

Apr - May 2026

React, FastAPI, WebSockets, SciPy, NetworkX, D3.js, Recharts, Docker

- Built a graph-based HVAC pressure-network solver using SciPy sparse linear algebra, then calculated duct velocity and volumetric flow from solved node pressures.
- Streamed solver convergence through WebSockets into a React dashboard with interactive network, pressure, velocity, residual, and tabular views.
- Added pytest benchmarks for pressure drop, branch behavior, flow conservation, topology convergence, and solver stability.

Crop Price Prediction - Senegal | *ML Engineer / Data App Developer | Local Streamlit App*

Nov - Dec 2025

Python, XGBoost, scikit-learn, TimeSeriesSplit, Streamlit, Plotly

- Engineered 83 temporal, lag, market, region, and crop features and evaluated models using chronological splitting and five-fold TimeSeriesSplit.
- Tuned XGBoost against Random Forest and achieved a reported test R2 of 0.981 on log-price predictions.
- Built a Streamlit interface with predicted XOF prices, 95% confidence intervals, confidence scoring, and Plotly forecasts.

RealiTech | *Sprint Lead / Product Owner | Live Hackathon Prototype*

Aug - Sep 2025

Node.js, Chatbot Workflows, Product Management, UX Design

- Led a one-week proptech sprint, shaped the roadmap, and coordinated buyer-assistance, developer tracking, agent routing, and post-sale product modules.
- Prototyped chatbot flows for loan pre-qualification, buyer profiling, financing guidance, and document checklist generation.

Mindspace | *Front-end Developer | Live Site*

Jan - Mar 2025

JavaScript, Accessibility, UX Design, Privacy-first Product Design

- Built a responsive, accessibility-minded student mental-health interface for resource discovery, guided meditation, moderated peer support, and crisis-help entry points.

AWARDS & RECOGNITION

- ASEAN Hackathon 2026 - Top 7, Smart Cities category, for IQueue; competition included 1,080+ participants, 240+ projects, 100+ academic institutions, and 9 ASEAN nations.
- TanUM - First Runner-Up, UPLB Innovation Olympics 3.0; 1st Place in Pitching and 2nd Place in Poster Competition, 2025 Mindanao-Wide Technology Transfer Conference.
- FIMiEx 2025, Malaysia - Gold award as an original developer and key contributor to TanimPro.
- HackEstate 2025 - Top UX Interface / Best UX recognition and Top 3 preliminary finish for RealiTech.

EDUCATION

Bachelor of Science in Computer Science | **University of Mindanao** | 2023 - Present

CERTIFICATIONS

- IT Specialist - Databases | Certiport
- IT Specialist - Java | Certiport
- CS50: Introduction to Computer Science | Harvard University

COMMUNITY LEADERSHIP

- Campus Torchbearer & Ambassador, Wadhwani Foundation Philippines | Nov 2024 - Present
- Programs Committee Member, Google Developer Group - Davao | Apr - Dec 2025
- Community Events Management Committee, Davao Startup Week | Jul - Sep 2025
- Communications Committee Member, DurianPy - Davao Python User Group | Jul - Oct 2025